

PAWS 2026: ALGEBRAIC GEOMETRY VIA COMPUTATION
PROBLEM SET 3

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Question 1: Proving Dickson's Lemma. You will prove a monomial ideal version of Dickson's Lemma by induction on the number of variables.

- (a) Why is Dickson's Lemma equivalent to the statement that every monomial ideal is finitely generated?
- (b) Prove Dickson's Lemma when $n = 1$.
- (c) Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be a monomial ideal. For $k \geq 0$ let

$$J_k = \left\{ f \in \mathbb{K}[x_1, \dots, x_{n-1}] \mid f x_n^k \in I \right\} \cup \{0\}.$$

Prove that J_k is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$ for all $k \geq 0$.

- (d) Show that the J_k form an ascending chain $J_0 \subseteq J_1 \subseteq J_2 \subseteq \dots$.
- (e) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$.
- (f) Assume that Dickson's Lemma holds in $n - 1$ variables. Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$.
- (g) For each $0 \leq k \leq N$, choose monomial generators $m_{k,1}, \dots, m_{k,r_k}$ for J_k . Via the inclusion $\mathbb{K}[x_1, \dots, x_{n-1}] \subset \mathbb{K}[x_1, \dots, x_n]$ view these as elements in $\mathbb{K}[x_1, \dots, x_n]$. Prove that I is generated by the finite set

$$\{m_{k,i} x_n^k \mid 0 \leq k \leq N, 1 \leq i \leq r_k\}.$$

- (h) Conclude Dickson's Lemma for all n .

Question 2: Existence of Gröbner Bases. Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and let $<$ be a monomial order.

- (a) Show that $\text{in}_{<}(I)$ is a monomial ideal.
- (b) By Dickson's Lemma, choose a minimal monomial generating set $m_1, \dots, m_t$ such that

$$\text{in}_{<}(I) = \langle m_1, \dots, m_t \rangle.$$

Show that there exist elements $g_1, \dots, g_t \in I$ such that $\text{LT}(g_i)$ divides m_i for all i .

- (c) Show that the minimality of $m_1, \dots, m_t$ implies $\text{LT}(g_i) = c_i m_i$ for some scalar $c_i \in \mathbb{K}^\times$.
- (d) Conclude that $\mathcal{G} = \{g_1, \dots, g_t\}$ is a Gröbner basis of I .

Question 3: The Circle–Parabola Cancellation, Now on Purpose. Work in $\mathbb{K}[x, y]$ with lex order $x > y$. Lecture 2 replaced the original circle–parabola equations by the equivalent list

$$g_1 = x^2 - y, \quad g_2 = y^2 + y - 1.$$

- (a) Recall that the S-polynomial of two nonzero polynomials f and g is

$$S(f, g) := \frac{\text{lcm}(\text{LM}(f), \text{LM}(g))}{\text{LT}(f)} f - \frac{\text{lcm}(\text{LM}(f), \text{LM}(g))}{\text{LT}(g)} g$$

Compute $S(g_1, g_2) = y^2 g_1 - x^2 g_2$ and reduce it to zero by (g_1, g_2) . Record the complete division identity, since it certifies that the remainder is zero.

- (b) We already proved in Lecture 2, by a special argument using the quotient, that (g_1, g_2) is a Gröbner basis. The calculation in part (a) suggests a general test. Before reading on, state the test you hope is true.
- (c) Suppose instead that an S -polynomial has nonzero remainder r . Explain why r belongs to the ideal but $\text{LM}(r)$ does not belong to the monomial ideal generated by the current leading monomials. What does this say about the current list?

Question 4: Why Buchberger's Algorithm Stops.

- (a) Prove that after every pass through the loop the current list G generates the same ideal as the input list.
- (b) Suppose a nonzero remainder r is adjoined. Why is $\text{LM}(r)$ outside the monomial ideal generated by the leading monomials in the old list? Conclude that this monomial ideal strictly increases.
- (c) If the current list G stops growing, it has only finitely many pairs, so the algorithm soon exhausts its list. Deduce that an infinite run would create an infinite strictly ascending chain of monomial ideals.
- (d) Apply the sequential form of Dickson's Lemma to the sequence of newly adjoined leading monomials and prove that this cannot happen. This is where Dickson's Lemma does its third job.
- (e) A pair which earlier had remainder zero still follows the same reduction after new divisors are appended. If its old remainder was a nonzero r , repeat the old reduction to reach r and then reduce by the appended divisor r itself. Thus every processed pair has remainder zero with respect to the final list. Once no unchecked pairs remain, apply Buchberger's criterion. State clearly which part of your argument proves termination and which proves correctness.

Question 5: Finding Infinity. This exercise will motivate the name "hyperplane at infinity". Work over $\mathbb{R}$ throughout.

- (a) Consider the standard affine chart

$$U_0 = \{[a_0 : a_1] \in \mathbb{P}_{\mathbb{R}}^1 \mid a_0 \neq 0\}.$$

Prove that every point of U_0 has a unique representative of the form $[1 : t]$. Use this to write down a bijection between $\mathbb{R}$ and U_0 .

- (b) Show that $\mathbb{P}_{\mathbb{R}}^1 \setminus U_0$ consists of exactly one point. We denote this point by ∞ . (Hint: If $a_0 = 0$, what must be true of a_1 , and how can you scale it?)
- (c) In projective coordinates, what point is ∞ ?
- (d) Now look at the other standard affine chart

$$U_1 = \{[a_0 : a_1] \in \mathbb{P}_{\mathbb{R}}^1 \mid a_1 \neq 0\}.$$

Show that every point of U_1 has a unique representative $[s : 1]$.

- (e) Verify the identity $[1 : t] = [\frac{1}{t} : 1]$ assuming $t \neq 0$. What happens as $t \rightarrow \pm\infty$?

Question 6: Getting Oriented in Projective Space. The definition is short, but the first few times one uses it, it is worth being pedantic.

- (a) Decide which of the following describe the same point in $\mathbb{P}_{\mathbb{Q}}^2$:

$$[1 : 2 : 3], \quad [2 : 4 : 6], \quad [1 : 2 : 4], \quad [0 : 2 : 3].$$

- (b) Why is $[0 : 0 : 0]$ not a point in $\mathbb{P}^2$?
- (c) Let $U_i \subseteq \mathbb{P}^n$ be the set where the i th coordinate is nonzero. Carefully show that U_i is a copy of $\mathbb{K}^n$ by normalizing the i th coordinate to 1.

- (d) Show that the collection $\{U_i\}$ for $i = 0, 1, \dots, n$ forms a cover of $\mathbb{P}^n$.
- (e) Let $a, b, c, d \in \mathbb{K}$, where a, b are not both zero and $c \neq d$. In $\mathbb{P}^2$ with homogeneous coordinates $[x : y : z]$, verify that $ax + by + cz$ is homogeneous.
- (f) Notice that the two affine lines $ax + by = c$ and $ax + by = d$ are parallel and hence never meet in $\mathbb{K}^2$. Consider their projective counterparts, $ax + by = cz$ and $ax + by = dz$. Show that the lines defined by these two homogeneous equations in $\mathbb{P}_{\mathbb{K}}^2$ intersect at $[b : -a : 0]$.
- (g) The point $[b : -a : 0]$ corresponds to a line through the origin of $\mathbb{K}^3$. Identify this line and explain how it relates to the two planes in $\mathbb{K}^3$ defined by $ax + by = cz$ and $ax + by = dz$.
- (h) Which part of the preceding calculation would break down if we tried to use a polynomial with terms of different degrees? Explain why homogeneity is necessary to make projective equations possible.

Question 7: The Naive Homogenization Fails. Let

$$I = \langle x^2 - y, xy - z \rangle \subseteq \mathbb{K}[x, y, z]$$

be the ideal defining the twisted cubic.

- (a) Homogenize the two displayed generators to obtain

$$x^2 - wy, \quad xy - wz.$$

Show that both vanish on the entire projective line

$$L = \{[w : x : y : z] \in \mathbb{P}^3 \mid w = x = 0\}.$$

Thus two equations which describe exactly the right affine points have quietly picked up a whole line of spurious points at infinity.

- (b) Compare this with the parametrization, which ought to know better. On the chart $s \neq 0$ of $\mathbb{P}^1$, write $u = t/s$. We want to recover the affine point $[1 : u : u^2 : u^3]$. Clear denominators so that all four coordinates become homogeneous of the same degree to obtain a map of the following form:

$$\nu_3 : \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \quad [s : t] \longmapsto [f_1(s, t) : f_2(s, t) : f_3(s, t) : f_4(s, t)].$$

Verify that the four coordinates have no common zero, so the map is well defined. Which single point appears when $s = 0$?

- (c) Which relation in the affine ideal is absent from our homogenized list, and how does Buchberger's algorithm discover it?

Question 8: The Twisted Cubic Finds Its Point at Infinity. Continue with the notation above, and let $J = \langle x^2 - wy, xy - xz, y^2 - xz \rangle \in \mathbb{K}[w, x, y, z]$.

- (a) Set $w = 0$ in the three generators of J . Show that the extraneous line from Exercise 7 is cut down to the single point $[0 : 0 : 0 : 1]$. Which equation does the cutting, and why could neither of the other two equations have done it?
- (b) Substitute

$$\nu_3([s : t]) = [s^3 : s^2t : st^2 : t^3]$$

into the three generators of J and show that they vanish. Conversely, show that every point of $\mathbb{V}_{\mathbb{P}^3}(J)$ lies in the image of ν_3 , treating $w \neq 0$ and $w = 0$ separately. Thus J cuts out precisely the parametrized curve, points at infinity included.

- (c) Check that $x^2 - y, xy - z, y^2 - xz$ is a Gröbner basis for the defining ideal of the twisted cubic, under the graded reverse lexicographic ordering.
- (d) Apply Theorem 15 from the notes for lecture 3, to the basis from the previous part. Check that its three homogenizations generate J , and conclude that $J = I^h$. Explain in one sentence why the original two generators carried no such guarantee.